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EDUCATION

B.S., Peking University, Beijing, China

Sep 2015 – July 2019

School of Earth and Space Sciences

Thesis Topic: Radiative Hydrodynamics Modeling of Eruptive Events on the Sun and M Dwarfs

Ph.D., University of Michigan, Ann Arbor, MI

Sep 2019 – Dec 2023

Department of Climate and Space Sciences and Engineering

Dissertation Topic: New Insights into the Coronal Heating Problem: Analysis of Spectral Line Widths in the Solar Corona

EXPERIENCE

Peking University, Beijing, China

Apr 2017 – July 2019

Undergraduate research supervised by Dr. Hui Tian

- Investigation of an S-shape Flare Ribbon with SDO and NVST observations
- Spectroscopic Diagnostics of IRIS data

National Solar Observatory, Boulder, CO

July 2018 – Sep 2018

Summer research supervised by Dr. Adam F. Kowalski

- Radiative Hydrodynamics Modeling of Mg II Lines at Solar Flare Ribbons

University of Michigan, Ann Arbor, MI

Sep 2019 – Dec 2023

Advisor: Dr. Enrico Landi

- Spectral Line Widths at Polar Off-limb Corona Observed by Hinode/EIS
- Ion Temperatures in a Polar Coronal Hole Observed by Hinode/EIS and SOHO/SUMER

Institute for Astronomy, University of Hawaii, Honolulu, HI

July 2022

Short visiting working with Dr. Shadia Habbal and Dr. Adalbert Ding

- Spectroscopic Observations during the 2017 Total Solar Eclipse

Physikalisch-Meteorologische Observatorium Davos/World Radiation Center (PMOD/WRC), Switzerland

Jan 2024 – Present

Postdoctoral Researcher supervised by Prof. Louise Harra

- Upflows at Active Region Boundaries

Institute for Particle Physics and Astrophysics, ETH Zürich

Jan 2025 – Present

Postdoctoral Researcher supervised by Prof. Louise Harra

- Upflows at Active Region Boundaries

PUBLICATIONS

- Zhu, Y.**, Harra, L. K., Barczynski, K., *et al.*, “Active region upflows in various coronal structures and their coupling to the lower atmosphere”, 2025, [A&A](#), **701**, A205
- Zhu, Y.**, Habbal, S. R., Adalbert, D., Yamashiro, B., Landi, E., Benjamin, B., Constantinou, S., Nassir, M., “Spectroscopic Observations of the Solar Corona during the 2017 August 21 Total Solar Eclipse: Comparison of Spectral Line Widths and Doppler Shifts between Open and Closed Magnetic Structures”, 2024, [ApJ](#), **966**, 122
- Zhu, Y.**, Szente, J., Landi, E., “Estimating Ion Temperatures at the Polar Coronal Hole Boundary”, 2023, [ApJ](#), **948**, 90
- Zhu, Y.**, Szente, J., Landi, E., “Fe XII and Fe XIII Line Widths in the Polar Off-limb Solar Corona up to 1.5 Solar Radii”, 2021, [ApJ](#), **913**, 74
- Zhu, Y.**, Kowalski, A. F., Tian, H., Uitenbroek, H. *et al.*, “Modeling Mg II h, k and Triplet Lines at Solar Flare Ribbons”, 2019, [ApJ](#), **879**, 19.
- Harra, L., Kusano, K., **Zhu, Y.**, *et al.*, “The Behaviour of the EUV Corona Before Flares in Regions with High Free Magnetic Energy”, 2026, [SoPh](#), **301**, 93
- Plowman, J. E., Hassler, D. M., Molnar, M. E., Shrivastav, A. K., Varesano, T., Auchère, F., Fludra, A., Kucera, T. A., Wang, T. J., **Zhu, Y.**, “A new method of deriving Doppler velocities for Solar Orbiter SPICE”, 2026, [A&A](#), **706**, A171
- Kontogiannis, I., **Zhu, Y.**, Barczynski, K., *et al.*, “Near-continuous tracking of solar active region NOAA 13664 over three solar rotations”, 2025, [A&A](#), **704**, A105.
- Barczynski, K., Janvier, M., Nelson, C. J., Schad, T., Tritschler, A., Harra L. K., Müller, D., Parenti, S., Valori, G., Cauzzi, G., **Zhu, Y.**, “First coordinated observations between Solar Orbiter and the Daniel K. Inouye Solar Telescope”, 2025, [A&A](#), **701**, A77.
- Huang, Z., Velli, M., Shi, Chen., **Zhu, Y.**, *et al.*, “Dominance of 2 Minute Oscillations near the Alfvén Surface”, 2024, [ApJL](#), **977**, 12.
- Yang, Z., Tian, H., **Zhu, Y.**, Xu, Y., Chen, L., Sun, Z., “Is It Possible to Detect Coronal Mass Ejections on Solar-type Stars through Extreme-ultraviolet Spectral Observations?”, 2024, [ApJ](#), **966**, 24.
- Yang, Z., Tian, H., Bai, X., Chen, Y., Guo, Y., **Zhu, Y.**, *et al.*, “Can We Detect Coronal Mass Ejections through Asymmetries of Sun-as-a-Star Extreme-ultraviolet Spectral Line Profiles?”, 2022, [ApJS](#), **260**, 36.
- Zhang, J., Tian, H., Solanki, S. K., Wang, H., Peter, H., Ahn, K., Xu, Y., **Zhu, Y.**, *et al.*, “Dark Structures in Sunspot Light Bridges”, 2018, [ApJ](#), **865**, 29
- Chen, Y., Tian, H., Su, Y., Qu, Z., Deng, L., Jibben, P. R., Yang, Z., Zhang, J., Samanta, T., He, J., Wang, L., **Zhu, Y.**, *et al.*, “Diagnosing the Magnetic Field Structure of a Coronal Cavity Observed during the 2017 Total Solar Eclipse”, 2018, [ApJ](#), **856**, 21
- Tian, H., Yurchyshyn, V., Peter, H., Solanki, S. K., Young, P. R., Ni, L., Cao, W., Ji, K., **Zhu, Y.**, *et al.*, “Frequently Occurring Reconnection Jets from Sunspot Light Bridges”, 2018, [ApJ](#), **854**, 92
- Song, Y. L., Guo, Y., Tian, H., Zhu, X. S., Zhang, M., **Zhu, Y.**, “Observations of a White-light Flare Associated with a Filament Eruption”, 2018, [ApJ](#), **854**, 64

ACADEMIC AWARD

Outstanding Student Presentation Award (OSPA), AGU 2022 Fall Meeting	2022
Outstanding Student Poster Award, 2 nd China-Europe Solar Physics Meeting	2019

ACADEMIC ACTIVITIES

Referee for

- Astronomy & Astrophysics (A&A) and The Astrophysical Journal (ApJ)

Teaching

- **Fall 2025, 26:** TA leading exercise classes of the Physics of Space Weather.

Talks

- “*Measuring Alfvén wave energy flux in closed and quasi-open coronal structures*”, 7th Swiss SCOSTEP Workshop Nov 2025
- “*Chromospheric Fibrils in Mossy Active Region Plage Observed by DKIST, IRIS, and Solar Orbiter*”, Hinode-18/IRIS-16 June 2025
- “*Upflows in a Decaying Active Region and its Potential Contribution to the Solar Wind*”, EGU 2025 April 2025
- “*Identifying Potential AR Upflow Regions in SPICE Doppler maps as Counterparts to Upflows Observed by Hinode/EIS: Preliminary Results with PSF Deconvolution Efforts*”, SPICE Consortium meeting June 2024
- “*Spectroscopic Observations of the Solar Corona in Different Wavelengths and from Multiple Vantage Points*”, Joint Solar Orbiter, Parker Solar Probe, and DKIST Meeting 2024 April 2024
- “*Spectroscopic Observations of the Corona in the Visible and Extreme Ultraviolet during the 2017 August 21 Total Solar Eclipse*”, SPD 2023 Aug 2023
- “*Spectroscopic Observations of the Corona in the Visible and Extreme Ultraviolet during the 2017 August 21 Total Solar Eclipse*”, AGU 2022 Fall Meeting Dec 2022
- “*Ion Temperature Diagnostics at the Off-limb Coronal Hole Boundary and Post-flare Loops*”, TESS 2022 Meeting Aug 2022
- “*Modeling Mg II h, k and Triplet Lines at Solar Flare Ribbons*”, ISSI/ISSI-BJ International Team Workshop: Diagnosing Heating Mechanism in Solar Flares Through Spectroscopic Observations of Flare Ribbons Oct 2018
- “*An S-shape Flare Ribbon Observed by NVST and SDO/AIA*”, The 2017 Annual Meeting of Chinese Geoscience Union (CGU) Oct 2017

Posters

- “*Measuring Alfvén wave energy flux in closed and quasi-open coronal structures*”, Solar Orbiter-IRIS-Aditya Joint Workshop Mar 2025
- “*Upflows with different morphologies at active region boundaries observed by Hinode and Solar Orbiter*”, ESPM-17 Sep 2024
- “*Heating of Heavy Ions in Active Regions Observed by SUMER: Flares, Loop brightening, and Coronal Rain*”, AGU 2023 Fall Meeting Dec 2023
- “*Spectroscopic Observations of the Solar Corona in the Visible, Extreme Ultraviolet, and Near-Infrared during the 2017 Total Solar Eclipse*”, Hinode-16/IRIS-13, Niigata, Japan Sep 2023
- “*Active Region Heating: Flares, Loop brightening, and Coronal Rain*”, SHINE Workshop 2023, Aug 2023
- “*Estimating Ion Temperatures at the Polar Coronal Hole Boundary*”, SHINE Workshop 2022, June 2022
- “*Ion Temperature Diagnostics at the Off-limb Coronal Hole Boundary and Post-flare Loops*”, AGU 2021 Fall Meeting Dec 2021

- “*Fe XII and Fe XIII Line Widths in the Polar Off-limb Solar Corona up to 1.5 Solar Radii*”, Hinode-14/IRIS-11 Meeting *Oct 2021*
- “*Fe XII and Fe XIII Line Widths in the Polar Off-limb Solar Corona up to 1.5 Solar Radii*”, SHINE 2021 *Jul 2021*
- “*Fe XII and Fe XIII Line Widths in the Polar Off-limb Solar Corona up to 1.5 Solar Radii*”, AGU 2020 Fall Meeting *Dec 2020*
- “*Modeling Mg II h, k, and Triplet Lines at Solar Flare Ribbons*”, The 2nd China-Europe Solar Physics Meeting *May 2019*
- “*An S-shape Flare Ribbon Observed by NVST and SDO/AIA*”, The 4th Chinese Space Weather Conference *Aug 2017*